Semantic Unsupervised Change Detection of Natural Land Cover With Multitemporal Object-Based Analysis on SAR Images

Donato Amitrano, *Member, IEEE*, Raffaella Guida, *Member, IEEE*, and Pasquale Iervolino, *Member, IEEE*

***Abstract*— Change detection is one of the most addressed topics in the remote sensing community. When performed on synthetic aperture radar images, the most critical issues are as follows: 1) the labeling of the identified changing patterns and 2) the scarce robustness of classic pixel-based approaches based on threshold segmentation of an appropriate change index, which tend to fail when multiple changes are present in the study area. In this work, a new methodology for unsupervised change detection in vegetation canopy is presented. It overcomes these limitations by exploiting multitemporal geographical object-based image analysis with the aim to make the intrinsic semantic of data emerge and direct the processing toward the identification of precise classes of changes through dictionary-based preclassification and fuzzy combination of class-specific information layers. The proposed methodology has been tested in ten different experiments covering agriculture and clear-cut deforestation applications. The results, validated against literature methods, highlighted the superiority of the proposed approach, which was quantitatively assessed in terms of standard classification quality parameters. On agriculture experiments, it allowed for an average increase in the detection accuracy of about 11% with respect to the best performing literature method, with an increment of the false alarm rate in the order of 0.5%. In case of deforestation, the registered detection accuracy was comparable to that achieved by the literature, while the most significant benefit was the reduction, of more than one-third, of the number of detected false deforestation patterns. Overall, the main characteristics of the proposed architecture are the robustness and the lack of any supervision, which makes it very well-suited for operational scenarios.**

***Index Terms*— Change detection, classification, multitemporal synthetic aperture radar (SAR), object-based image analysis.**

I. Introduction

THE detection of landscape changes due to human activities or natural events was one of the most addressed topics in the remote sensing literature [1]. It can be defined as the identification of the differences in the state of an object or of a pattern by observing it at different times [2]. This principle can be applied in diverse applications like, among the others, flood mapping [3]–[5], large displacements estimation [6]–[8], monitoring of shifting land use [9]–[12], urbanization [13]–[15], deforestation [16]–[20], and damage assessment after natural disasters [21]–[23]. Changes are highlighted by means of some temporal indicators which are then processed for the extraction of meaningful changing patterns.

In synthetic aperture radar (SAR) research, change detection problems typically cover two aspects, that is, the change indicator and the segmentation [24]. As for the former, there is a wide literature proposing methods, mainly based on multispectral (MS) data, exploiting the difference operator (see [25], [26]). However, as reported by Bovolo and Bruzzone [27], the probability distribution function (pdf) of the difference between two intensity images, assumed to be Gamma-distributed due to multilook, leads to different change detection accuracy in high-intensity regions compared with the low-intensity ones. Therefore, since the early 90's, the ratio operator was one of the most adopted solutions because its distribution depends only on the relative change in the average intensity response of the two images. This makes changes detected in the same way in both high- and low-intensity areas [28] and the detection more robust with respect to speckle and calibration inaccuracies [29]. Typically, the ratio image is expressed in logarithmic scale to make the distribution of the classes of interest more symmetrical and to enhance the differences between changed and unchanged pixels [29]–[31]. Other change indicators introduced in the past literature include likelihood ratios [10], [32]–[34], normalized multitemporal band comparison [35], [36], and information theoretical similarity measures such as the Kullback–Leibler divergence (KLD) [37] and the mutual information [38].

Once the change indicator is selected, the problem moves to the decision level. This was usually addressed with threshold segmentation. However, as reported in literature, effective thresholding of the identified information layer may be critical [38]. Basic algorithms based on histogram analysis, such as the Otsu thresholding [39], are likely to fail since they require: 1) a bimodal distribution in the input image and 2) the amounts of changed and unchanged pixels are comparable [40], which is a situation not feasible in operative scenarios. The same applies to empirical trial-and-error thresholding [28],

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Donato Amitrano is with the Department of Electrical and Information Technology Engineering, University of Napoli Federico II, 80125 Naples, Italy (e-mail: donato.amitrano@gmail.com).

Raffaella Guida is with the Ingegneria Elettronica e delle Telecomunicazioni, Università di Napoli, 80125 Naples, Italy.

Pasquale Iervolino is with the Department of Electronic Engineering, Surrey Space Centre, Guildford GU2 7XH, U.K.

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which requires strong supervision and is biased by the expertise of the operator, or for semi-supervised clustering [41], which relies on the availability of relevant training samples.

Bayes theory and the Kittler–Illingworth minimum-error thresholding have been exploited for strengthening the automatic thresholding algorithms [29], [42]–[44], which still represent the most adopted solutions in the literature. However, the underlying weakness of this approach is in the concept of the threshold itself, which is intrinsically not robust. It typically works well in the presence of abrupt and extreme changes, due to, as an example, floods or growing urban areas, but is not applicable, as an example, when there is the necessity of labeling multiple changes in the study area. Let us consider a situation in which there are more than one change in the scene, like a flood in agricultural area. At microwaves, the areas affected by harvesting or flood are both characterized by a decrease in the backscattering coefficient due to, in the first case, the lack of volumetric scattering contributions from the vegetation canopy [45] and, in the second, the change in surface roughness caused by the presence of a tick water layer over the usual soil [46]. Therefore, if the objective is the identification of changes in crops, then two thresholds are necessary: one isolating the changed pixels from the unchanged background and another segmenting agricultural land from flooded areas. Most of the analyzed literature cannot cope with this problem since thresholding is typically blind with respect to the type of changes.

This makes thresholding not suitable for operative scenarios, so much so that the major services based on change detection approaches, like the DETER [47] and the PRODES [48] for deforestation monitoring in the Amazon, or the European CORINE land cover change program [49], are based on the exploitation of MS images.

The objective of this work is the introduction of a new architecture exploiting the SAR data for the detection and labeling of changes in natural land cover, with special focus on those due to an increase or decrease in the biomass content in the resolution cell related to agricultural activities or clear-cut deforestation. The proposed methodology takes advantages of consolidated SAR techniques (especially in the preprocessing of the input time series) and introduces the main novelties in the information extraction process, which benefits new input products [50], [51] and concepts bridging traditional SAR data processing with modern geographical object-based image analysis (GEOBIA) [52] exploiting the intrinsic semantics of data.

The novelty here introduced can be summarized as follows. Unlike the past literature, in which the segmentation is implemented using statistical principles often unable to interpret the nature of changes, here the process is initialized using a relevant dictionary of attributes. This allows for directing the processing toward the identification of the classes of interest, avoiding the confusion with phenomena having similar characteristics on SAR images. The dictionary, build upon semantic data clustering, acts as a preclassifier [53]. Then, two information layers, one scattering-based and one texture-based [54], are used to feed a fuzzy classification system which is able to erode the over dimensioned preclassification map and to assign relevant labels to patterns of canopy removal and growth in a fully automatic way. All this constitutes a novel change detection paradigm with respect to the past literature, which mainly relies on thresholding. The experimental results and the comparison with the reference SAR change detection literature show that the introduction of a semantic information layer allows for separating different phenomena having the same behavior in the backscattering domain, thus increasing the robustness of the detection system, even in critical scenarios. This reflects in a substantial improvement of the classification quality parameters, which outperform those from the past literature.

This article is organized as follows. The proposed methodology is presented in Section II, along with validation data sets and reference literature. The experimental results and comparison against the literature are discussed in Section III. Conclusions are drawn at the end of the work.

II. Data, Proposed Methodology, and Reference Literature

A. Data

The proposed methodology has been tested on five data sets addressing both standard scenarios, with just one dominating change due to biomass decrease or increase as a consequence of agricultural cycles, and critical ones including the presence of phenomena (like floods), special surfaces (such as sands), or geometric distortions (i.e., layover and shadowing) which can be source of misclassifications. The reference data for validation are derived from NDVI maps extracted from the closest in time cloud-free MS acquisitions with respect to the extremes of the SAR observation period. The rationale for the building of the reference data is sketched in Fig. 1. Starting from an (ideally) fully developed vegetation canopy, the biomass decrease can be associated with harvesting or the subsequent partial regrowth (or more generally to the presence of a vegetation canopy less developed compared with time zero situation). Therefore, the reference data for this class include changes due to the passage from t_0 to t_1 or from t_0 to t_2, depending on the state of the particular crop field at the beginning and the end of the SAR time series. In an SAR image, both the cases correspond to a vegetation canopy removal mechanism. Therefore, in the following, the biomass decrease class will be generally referred to in this way.

Conversely, biomass increase is due to the growth of vegetation, either starting from a bare soil or from an already developed vegetation canopy. Thus, relevant reference data include changes due to the passage from t_0 to t_1 or from t_0 to t_2, depending on the state of the particular crop field at the beginning and the end of the SAR time series. The biomass increase is in any case related to the growth of plants. Therefore, in the following, this class will be referred to as vegetation canopy growth. The reference data do not include phenomena related, as an example, to the health of forest canopies that can have effects on the NDVI [55] but are scarcely detectable using microwaves, which are typically less sensitive to biomass variations [56] especially at the wavelengths of the most available sensors.

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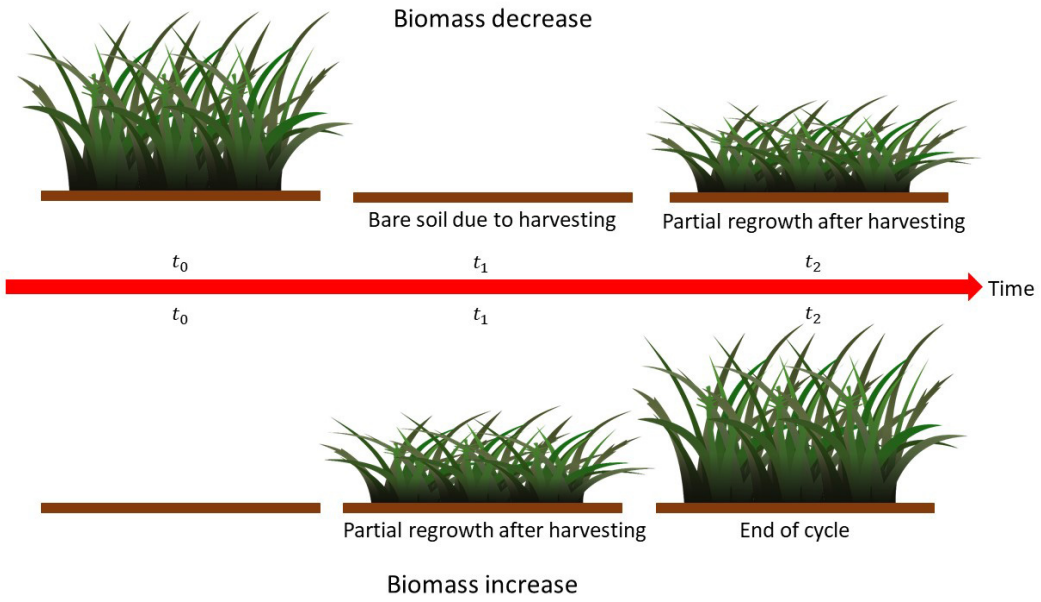


Fig. 1. Reference phenomenology for the derivation of validation data from MS images. (Top) Biomass decrease can be due to total harvesting or partial regrowth after it. (Bottom) Biomass increase can be due to partial regrowth starting from bare soil or further growth starting from a less developed vegetation canopy.

Back to the analyzed data sets, standard situations (i.e., scenes with one dominant change) are considered at first. They concern rural areas in Vietnam and Cambodia. The Vietnam data set covers an area of about 1800 km^2 (4823 $\times$ 3800 pixels) surrounding the city of My Phuóc. It is composed of 58 Sentinel-1 ground range detected (GRD) images with 10-m pixel spacing acquired in interferometric wide swath mode between March 13, 2018 and March 2, 2019. This data set represents a mixed environment, characterized by dominant agricultural lands, scattered permanent vegetation, water surfaces, and small urban agglomerates. The reference data for this experiment were derived from Sentinel-2 MS Level-2A images (i.e., Sen2Cor atmospheric correction [57] was applied) acquired in the closest cloud-free available dates with respect to the start and end date of the aforementioned observation period. In particular, in this case, images acquired on March 12, 2018, and March 2, 2019, were used. The procedure for the retrieval of reference data was based on the temporal analysis of NDVI patterns. Vegetation canopy removal and growth patterns were identified by trial-and-error thresholding (supported by visual assessment) of the NDVI difference map. In case of vegetation canopy removal, the detection aimed at identifying areas characterized by high NDVI in the first acquisition followed by a significant drop in the second one. Conversely, the condition for growth was set as NDVI in the first acquisition followed by an increment in the second one. The threshold on the NDVI difference map for the identification of reference removal and growth patterns was set at ± 0.3. This value has been used to generate reference data for all the data sets.

The Cambodia data set [see Fig. 2(a)] covering an area of about 9000 km^2 (12 503 $\times$ 7257 pixels) is composed of 32 Sentinel-1 GRD images with 10-m pixel spacing acquired in interferometric wide swath mode between March 19, 2018, and March 26, 2019, over the city of Krong Strung Treng. This scene is fully rural with almost total absence of urban scatterers. The reference data for this data set were derived from a pair of Sentinel-2 Level-2A images acquired on March 12, 2018, and March 27, 2019, and analyzed for NDVI drops or increases as discussed above.

The third scene deals with a nonstandard scenario created with a collage (with final dimensions of 6000 $\times$ 6000 pixels) of four SAR data sets (each one composed of 22 images), as shown in Fig. 2(b). The scene placed in the upper right quadrant is the aforementioned Vietnam one. The bottom right quadrant represents the area of Doha (Qatar), thus characterized by significant presence of urban scatterers and sands. The bottom left quadrant reports a mountainous area close by the city of Xi'an (China), with dominant layover and shadowing features. The upper left quadrant depicts a rural area in the Brazilian Amazon nearby Ji Parana' village. Here, there is dominant natural land cover, in the form of both forests and crops. The changes due to canopy removal are located in the Vietnam scene, those relevant to canopy growth in the Vietnam and Brazil ones. Data for this last data set have been acquired by the Sentinel-1 sensor between August 3, 2017, and May 31, 2019. Images are in the GRD format and have 10-m spatial resolution. Reference growth patterns from this data set have been estimated from Sentinel-2 Level-2A images acquired on July 27, 2017, and May 28, 2019.

The fourth data set covers an area of about 3000 km^2 (5334 $\times$ 5630 pixels). It presents the situation of two dominant changes in the study area, which is relevant to the Albufera natural reserve (near Valencia, Spain). This area is characterized by the presence of rice cultivations within a protected environment [58]. Crops are periodically flooded [see Fig. 2(c)] according to the conditions required for the preservation of the species populating the area. Overall, the landscape

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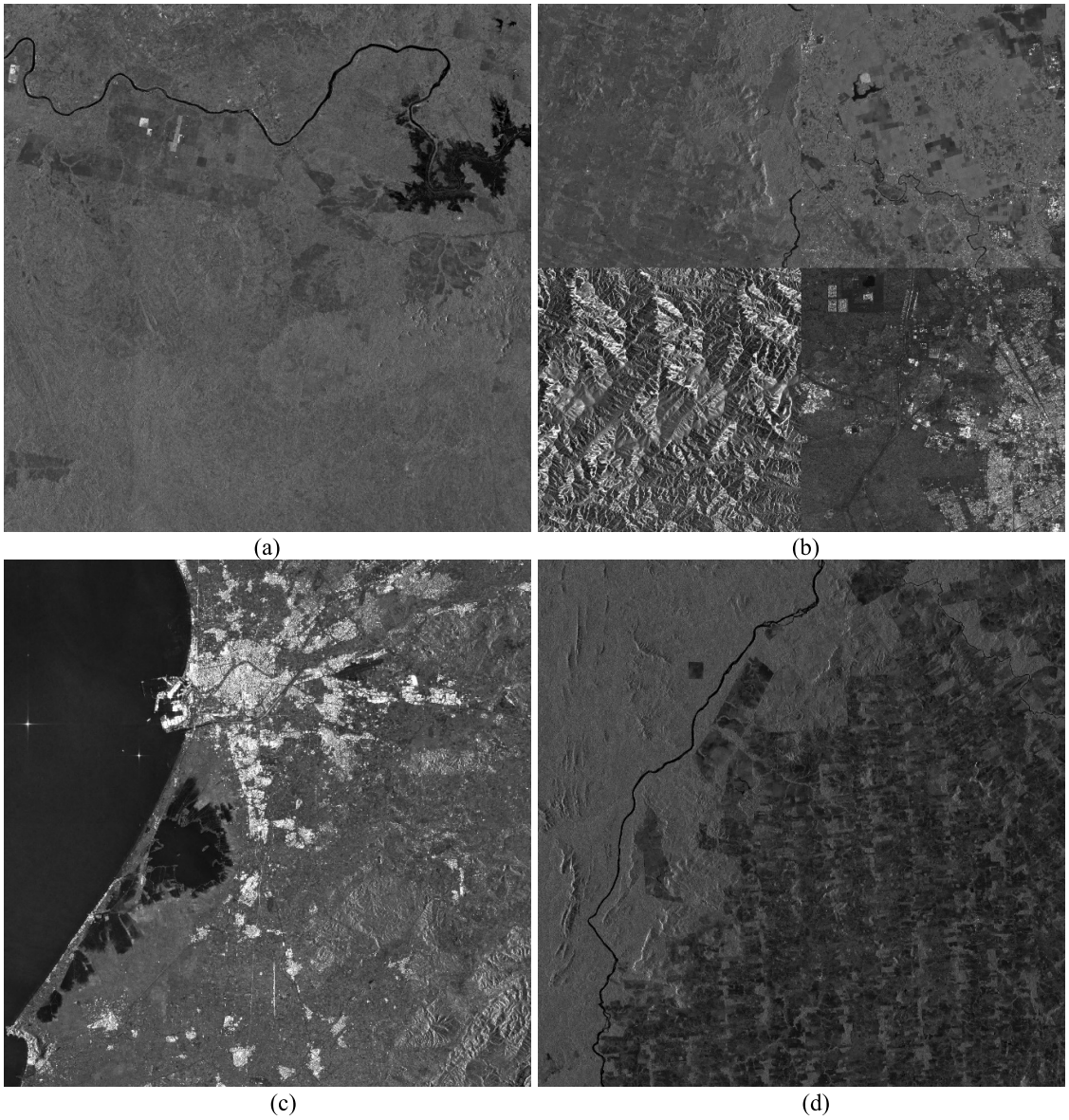


Fig. 2. Four out of five available data sets. (a) Cambodia. (b) Vietnam–Qatar–China–Brazil collage. (c) Albufera natural reserve. (d) Simulated clear-cut deforestation pattern in the Brazilian Amazon. Images are in the intensity format.

is mixed, with the presence of large and small urban agglomerates, water surfaces, and agricultural land. This data set is constituted by 22 Sentinel-1 GRD images acquired between April 6, 2017, and December 14, 2017, with 10-m spatial resolution. Reference canopy removal data have been collected starting from Sentinel-2 Level-2A images acquired on June 4, 2017, and December 17, 2017.

The last data set [see Fig. 2(d)] is the Brazil one previously introduced (its total extension is of about 6000 km^2, i.e., 7653 $\times$ 7737 pixels) which has been investigated for a subclass of the canopy removal pattern, that is, clear-cut deforestation. At microwaves, as explained in Section I, these two phenomena are both characterized by a decrease in the total backscattering due to the lack of volumetric scattering contribution as an effect of the removal of vegetation canopy. Although they are both impulsive events, it is easy to understand that the removal of vegetation due to harvesting and that due to clear-cut of trees are different phenomena occurring at different time scales. Deforestation is a process that can take several years before showing significant patterns at the medium resolution (from 10 to 30 m) of the most available free data (basically those coming from the Landsat and the Copernicus missions). Therefore, its patterns have been simulated by inserting, in an image dated in the middle of the temporal window of the time series, a pattern of about 100 ha of bare soil in an area of undisturbed forest. The pattern was extracted from one of the other available data set [see the dark pattern left to the river in the upper part of Fig. 2(d)]. This experiment is useful in understanding the degree of separability of the class deforestation from that of the agricultural land.

Information about available SAR and MS data are summarized in Table I.

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TABLE I
SUMMARY OF THE AVAILABLE DATA

Dataset	SAR images	SAR start	SAR end	MS start	MS end	Target class
Vietnam	58	13/03/2018	02/03/2019	12/03/2018	02/03/2019	Canopy removal Canopy growth
Cambodia	32	19/03/2018	26/03/2019	12/03/2018	27/03/2019	Canopy removal Canopy growth
Albufera	22	04/06/2017	14/12/2017	04/06/2017	17/12/2017	Canopy removal
Brazil	59	03/08/2017	31/05/2019	27/07/2017	18/05/2019	Deforestation

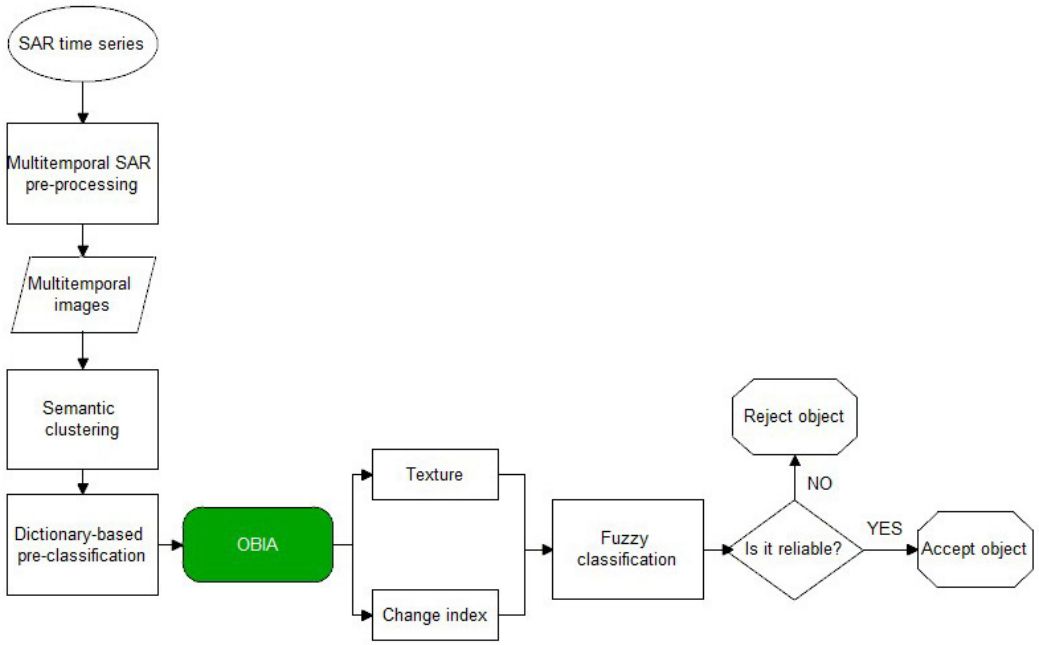


Fig. 3. Block diagram of the proposed methodology.

B. Proposed Methodology

The block diagram of the proposed methodology is depicted in Fig. 3. It receives in input a time series of SAR images, initially entering a standard multitemporal preprocessing block in which calibration, coregistration, and despeckling (see [51] for further details) are performed. In particular, multitemporal De Grandi despeckling [59] is here exploited. This filter is able to properly handle the variability of the scene, restituting the filtered reflectivity function for each of the N images composing the time series, irrespectively of the temporal changes in the local land cover.

At the end of the preprocessing phase, all the images composing the data set are geometrically and radiometrically comparable, that is, generic pixels having coordinates (i,j) in the matrix of the SAR sigma nought represents, in all the images of the series, the same portion of terrain and shows the same reflectivity value if the scene is unchanged. In Fig. 4(a), one of the available images belonging to the Vietnam data set after multitemporal preprocessing is shown.

Data are then combined in multitemporal change-detection-oriented color composites as described in [13]. The objective is to provide a picture of the changes occurred in the imaged scene with respect to a reference situation for the application at issue. For the sake of clarity, let us consider the image reported in Fig. 4(b). It has been obtained by loading a reference image (acquired on March 13, 2018) on the red band, a test image on the green band, and a texture layer on the blue band. The test image has been acquired on February 3, 2019. It represents the time at which changes have to be evaluated.

In case of vegetation canopy removal, the pattern to be identified is that of something (i.e., crops) that has been removed from the scene. Therefore, a reduction of the backscattering in the test image, compared with that of the reference one, is expected. This means that the deforested areas should appear in reddish color in the composition, also because the texture (and therefore the contribution of the blue band) is expected to be low for bare soils.

On the other hand, growing vegetation contributes to the increase in the total scene return because of the larger volumetric backscattering it is responsible for [45]. Therefore, these areas are rendered with a dominant green color in the composition reported in Fig. 4(b) [51].

If the two intensity layers are balanced, it means that the scene is unchanged in that area within the given temporal frame. This is the case of the lake in the top right part of the scene or of permanent vegetation spreading all over the scene. In the first case, due to lower backscattering characteristics of water, the composition returns black. In the second, it turns to be a yellowish shade.

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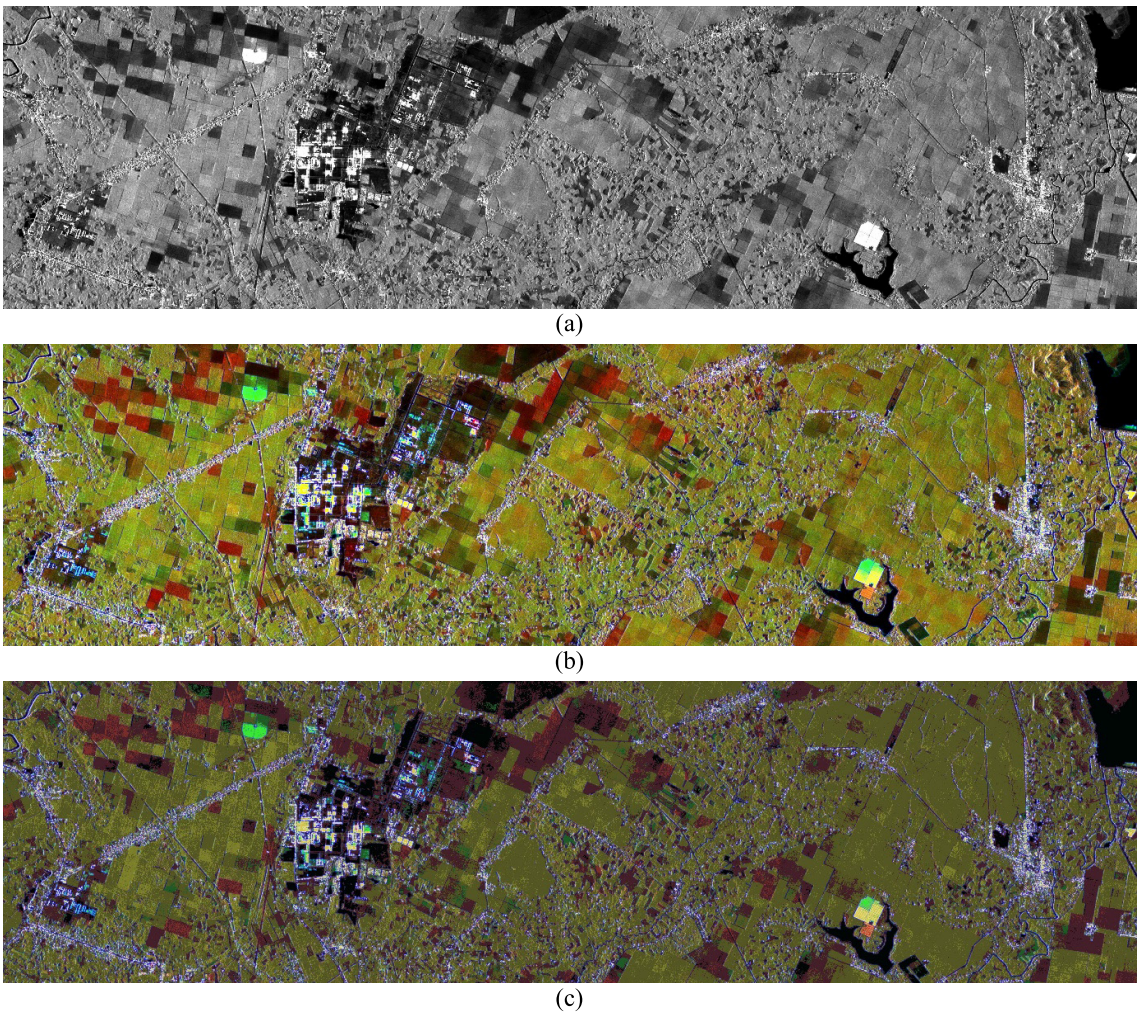


Fig. 4. Vietnam data set. (a) Sentinel-1 SAR image acquired on March 13, 2018, near the city of My Phuóc. (b) Multitemporal change-detection-oriented composite (test image acquired on February 3, 2019). (c) 64-class semantic clustering of the temporal color composite. Each cluster is attached with a semantic attribute referring to its color.

The texture channel (blue band) is represented by the data range, which is defined as follows [13]:

$$\mathrm{d_r} = \max i_w - \min i_w, \quad \mathrm{d_r} \in [0, 255] \tag{1}$$

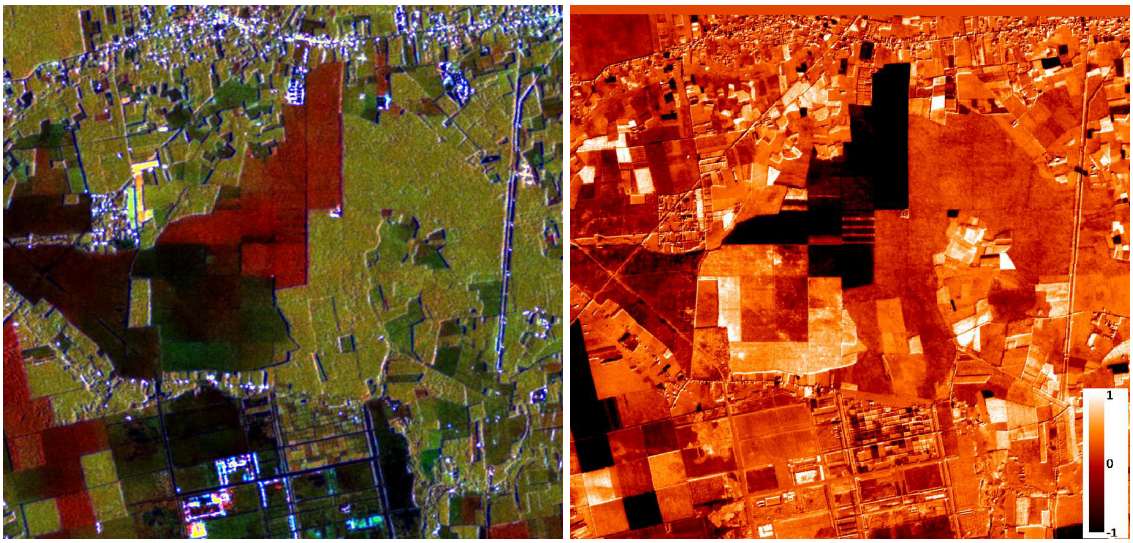
where i_w is the intensity of the generic pixel falling in a window w sliding on the test image. The texture is expected to be relevant in urban agglomerates, which are typically characterized by a mixed presence of highly reflective man-made targets (i.e., buildings) and low backscattering features (such as roads and shadows). The raising of a significant texture turns the composition to blue and white shades on these areas. As indicated in [13], the texture channel used in the RGB composition is relevant to the test image, that is, the green band of the composite. As suggested in [13], a calculation window of 50×50 m^2 has been used to produce this information layer.

In Fig. 5, a comparison between the Vietnam RGB composite and the NDVI difference map is shown. The reader should note that the red clusters in the first product correspond to the dark areas (representative of significant drop in the NDVI) in the second one. Conversely, areas exhibiting a positive change in the NDVI difference correspond to clusters with a more significant contribution of the green color in the RGB frame.

The multitemporal change detection color composite is ingested into a standard self-organizing map (SOM) clustering [60] modified to fit with the particular input data and to attach a relevant semantics to the output clusters. In the classic Kohonen's scheme, the objective is to map the input data from a high-dimensional space S to a space $\hat{S}$ having a limited number of elements, coinciding with the number of SOM neurons. In our case, the space S is represented by the RGB color space, whose dimension is [256 $\times$ 256 $\times$ 256], while the target $\hat{S}$ is a subset of the HTML color database, providing color labels useful to query the output cluster map in the feature space. The Euclidean distance between the SOM neurons and the database elements is computed to determine which of them belong to the SOM relevant with the image at hand. In particular, for each SOM neuron, the closest color code (and its label) within the HTML color database is picked for the association (see [53] for further details).

In Fig. 4(c), the output of the SOM algorithm for the image depicted in Fig. 3(b) is reported. This cluster map has two important characteristics: 1) its appearance being very similar to that of the input product, the semantic transfer from one product to the other is immediate and 2) each

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(a) (b)

Fig. 5. Comparison between the (a) Vietnam RGB composite and (b) NDVI difference map.

cluster is associated with a verbal attribute referring to its color.

As explained above, the color of the composition being linked to a specific physical process [51], the aforementioned verbal attribute can be used to identify the clusters relevant to, as an example, vegetation canopy removal (reddish clusters) or growth (greenish clusters). This identification is based on a prebuilt dictionary including all the words, that is, the colors included in the aforementioned HTML color database, linked to the phenomenon under analysis and represents the prior knowledge of the system concerning the mapping of SAR scattering mechanisms into the temporal color composite [53]. The selection of such words shall be made by an expert operator and is essential to teach the preclassifier for the identification of a specific phenomenon. This operation is due just once and represents the only interaction the user is required to have with the classification system.

Not all the words included in the dictionary have the same reliability regarding a specific phenomenon, since the same world can be associated with more phenomena. As an example, an increase in the backscattering can be due to an increase in the biomass within the same resolution cell and also to an increased percentage of humidity in the bare soil [61]. In the same way, a cluster located at the edge of a deforested area can exhibit a mixed response made of contributions of bare soils and canopies. Therefore, to minimize misclassifications, the dictionary is split into two parts, one reliable and one unreliable. The reliable dictionary includes the words which are more likely to be connected to the specific phenomenon. As an example, as explained above, vegetation canopy removal is expected to be represented with clusters having strong dominance of red. Therefore, the different shades of the color red are listed as belonging to the reliable dictionary. On the other hand, clusters exhibiting more balance between the two intensity channels (i.e., having a yellowish color) are considered unreliable and left aside in the first iteration of the OBIA processing block.

In Fig. 6, the downstream part of the workflow reported in Fig. 3, with an expanded view of the implemented OBIA, is reported. The stream comes from the semantic clustering and the dictionary-based preclassification, which is based on the reliable dictionary only. The relevant clusters are analyzed for average texture and scattering characteristics. The texture layer is represented by the data range texture discussed above [see (1)]. The scattering layer is represented by the change index introduced in [3] and [35] and defined as follows:

$$\mathrm{CI} = \frac{i_{t_1} - i_{t_2}}{i_{t_1} + i_{t_2}}, \quad \mathrm{CI} \in [-1, 1] \tag{2}$$

where i_{t_2} and i_{t_1} represent the intensity of the generic pixel at time t_2 and t_1, respectively. The change index is expected to be high where significant changes on the scene (like flood or deforestation) occurred, close to zero otherwise.

In Fig. 7, the information layers to be ingested into the fuzzy classification system are shown. In particular, Fig. 7(a) depicts the data range texture layer, while Fig. 7(b) reports the change index scattering layer. The comparison between these layers and the input color composite depicted in Fig. 4(b) makes evident the correspondence between candidate areas interested by vegetation canopy removal (i.e., those rendered in reddish color) and dark and bright regions in the texture and scattering layer, respectively. This means that as expected, bare soils (after the removal of vegetation cover) exhibit low texture and high change index. However, the reader should note that even the two lakes present in the scene (in the top and bottom right corners of the image) present similar properties, since water surfaces are typically characterized by low texture and can be affected by significant variations in the backscattering due to Bragg scattering triggered by wind-induced surface roughness [62]. This means that by applying standard thresholding methods to these information layers, like in most of the analyzed literature, these objects are likely to be misclassified as areas interested by vegetation cover removal. Using the proposed approach, instead,

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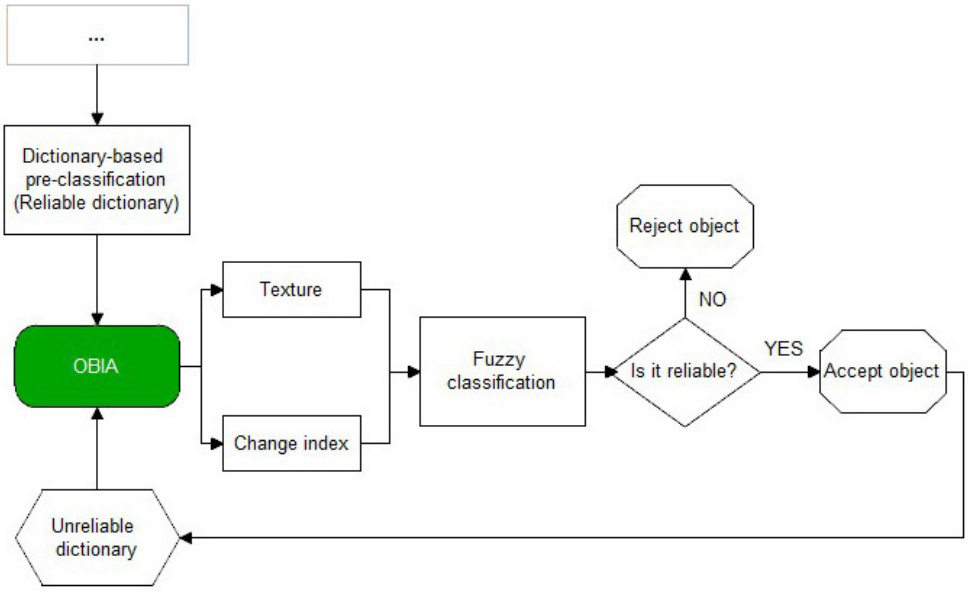


Fig. 6. Expanded view of the OBIA processing block.

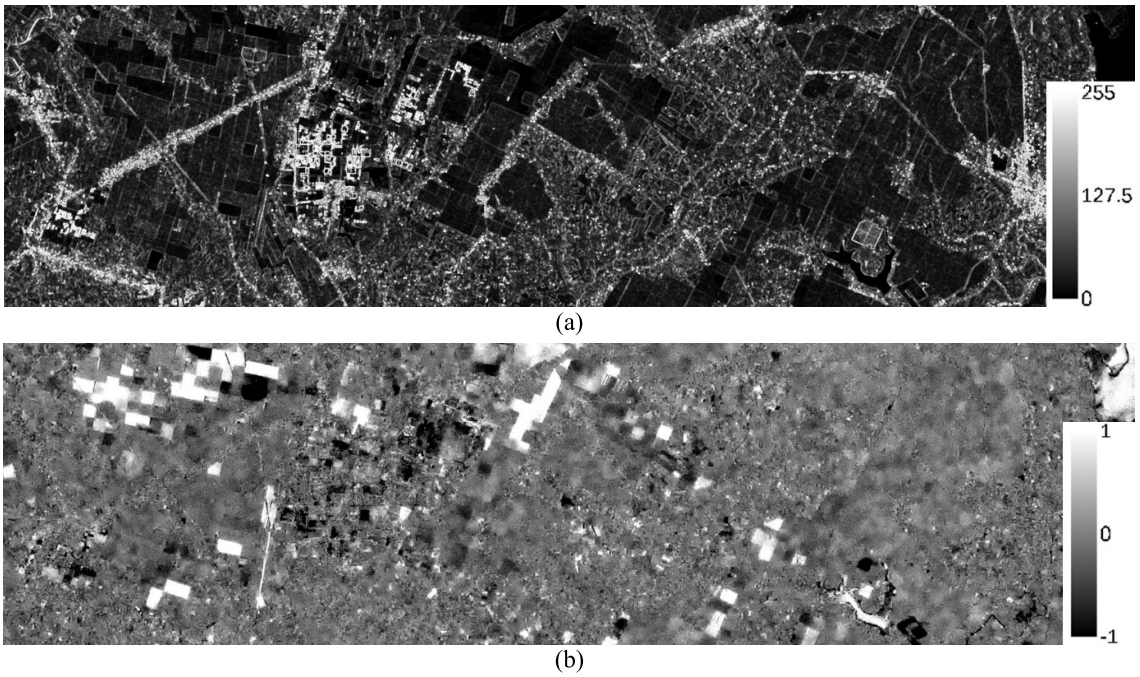


Fig. 7. Vietnam data set. (a) Data range texture layer. (b) Change index scattering layer.

these features are automatically discarded from the classification process, since the corresponding colors within the temporal composite do not belong neither to the reliable dictionary nor to the unreliable one.

The texture and scattering layer feed the fuzzy classifier for the assignment of the class vegetation canopy removal or growth. The reader should remember that at this point in the processing chain, the system is using just objects derived from the reliable dictionary.

Fuzzy systems represent an effective tool for managing the uncertainty connected with data [63] and have been widely exploited in various remote sensing applications [64]–[66] including change detection [67]. Input variables are modeled through membership functions corresponding to verbal attributes referring to their value. This makes the output classes (i.e., canopy removal or canopy growth) now identified by a statement and not by a threshold anymore. As an example, in the case of the canopy removal class, the statement is "The canopy removal class is characterized by high change index and low texture." The attributes "low" and "high" are modeled with appropriate membership functions (or fuzzy sets) which, usually, assume the form of S-type, Z-type, or π-type function.

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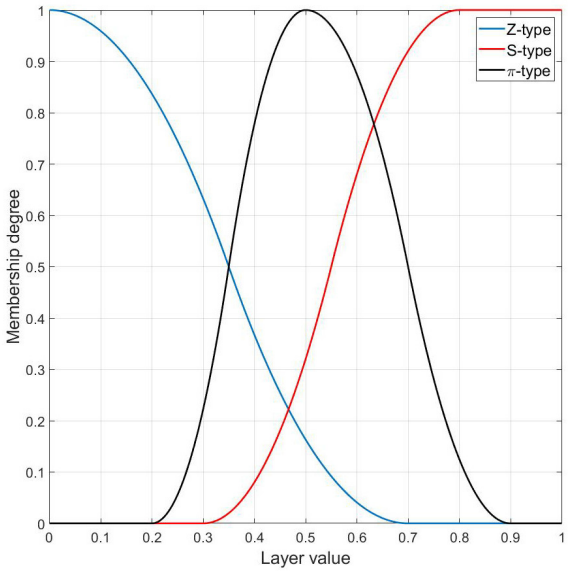


Fig. 8. Sample fuzzy sets. S-type (red curve), Z-type (blue curve), and π-type (black curve). The S- and Z-type membership functions are typically used to model the attributes "low" and "high," respectively. The π-type function (also known as bell-shaped) can be exploited to model intermediate behaviors associated with the attribute "average."

In particular, the S-type fuzzy set is defined as [68]

$$S(\mathrm{DN},a,b,c)=\begin{cases}0, & DN\le a\\ 2\left[\dfrac{\mathrm{DN}-a}{c-a}\right]^2, & a<DN\le b\\ 1-2\left[\dfrac{\mathrm{DN}-c}{c-a}\right]^2, & b<DN\le c\\ 1, & DN>c\end{cases} \tag{3}$$

in which DN means the digital number and a, c, and $b=(a+c)/2$ are the user-defined parameters of the fuzzy set. The Z-type function is obtained from the S-type one through relation $\mathrm{Z(DN},\ a,\ b,\ c)=1-\mathrm{S(DN},\ a,\ b,\ c)$. Finally, the π-type function is obtained by the junction of an S-type and Z-type functions sharing the parameters c and a, respectively. Sample S-type, Z-type, and π-type fuzzy sets are reported in Fig. 8 for the ease of the reader.

In this work, the attributes "low" and "high" are modeled differently for the two input layers. In particular, the attribute "low" for both the change index and the texture is modeled with a Z-type fuzzy set. The attribute "high" for the texture layer is given by an S-type membership function. That for the change index is modeled through a π-type fuzzy set because when the change is very high, it is likely to be due to flood rather than vegetation canopy removal. In other words, the change index is expected to be higher in flooded areas when compared with crop fields subject to harvesting because the reduction in the backscattering is typically higher when the bare soil is covered with a water layer than when the bare soil is affected by the removal of vegetation [46]. This phenomenology is well-represented by the π-type membership function assigning an increasing probability to changing patterns to be associated with vegetation canopy removal up to a certain value of the changing index. Then the function switches in its descending membership part, where the change index is more likely to be associated with floods. The experimental parameter setup for the above described fuzzy sets is provided in Table II.

TABLE II
EXPERIMENTAL SETUP FOR THE FUZZY SET DEFINED FOR EACH INFORMATION LAYER USED FOR CHANGES IDENTIFICATION

Information layer	Attribute	Fuzzy set	Params
Change index	Low	Z-type	a = 0, c = 0.25
	High	$\pi-\text{type}\begin{cases}\text{S}-\text{type}\\ \text{Z}-\text{type}\end{cases}$	a = 0.1, c = 0.5 a = 0.5, c = 0.8
Texture index	Low	Z-type	a = 0, c = 0.4
	High	S-type	a = 0.2, c = 0.6

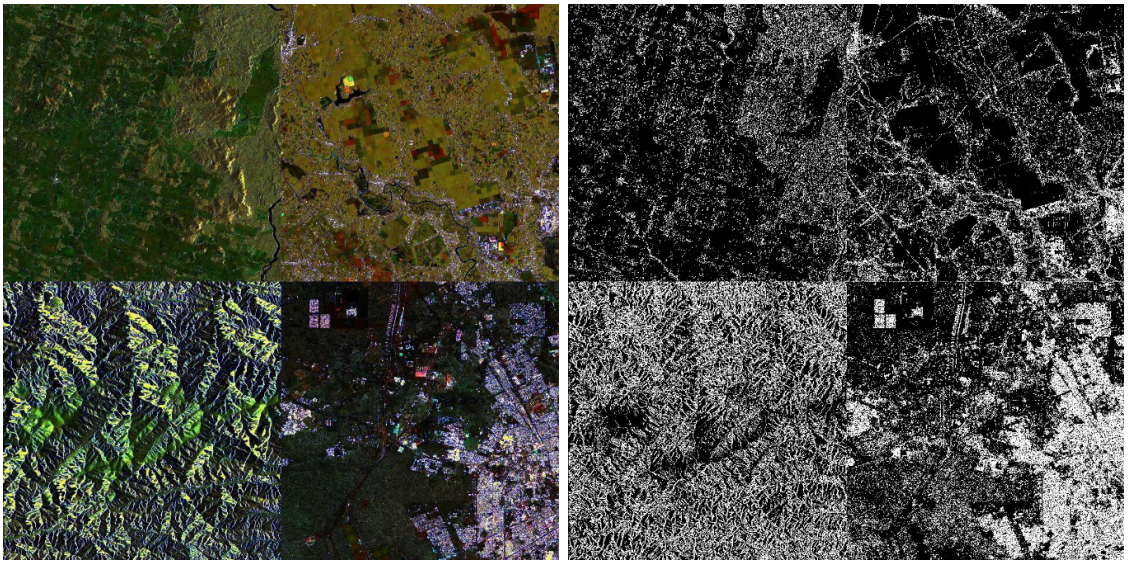
The above-described fuzzy system outputs four different classes, that is, those characterized by "high" change index and "low" texture, "high" change index and "high" texture, "low" change index and "high" texture, and "low" change index and low "texture." Each image segment, preclassified using the reliable dictionary, exhibits a membership degree to each class but, among them, just the first combination of attributes (i.e., "high" change index and "low" texture) respects the phenomenology of, for instance, vegetation canopy removal. The defuzzification of this soft classification map is made with the maximum membership criterion [69]. Each region is assigned to the class exhibiting the highest membership degree. For the purpose of the identification of vegetation canopy removal patterns, only those having the maximum membership corresponding to the class "high" change index and "low" texture are kept and sent to the unreliable dictionary loop.

In this loop, relevant clusters are added one by one, starting from that having the highest average change index, to the core identified by the objects accepted during the reliable dictionary filtering. This means that as the loop moves deeper within the unreliable dictionary, the core changing patterns will be aggregated to regions less and less likely to represent areas of vegetation canopy alteration due to the lower tendency to change in the associated clusters.

The acceptance/rejection mechanism of the new candidate changing patterns (see lower part of the diagram reported in Fig. 6) is the same as the one just discussed. The only modification is implemented in the fuzzy classification system in the modeling of the verbal attribute "low" and "high." When dealing with the unreliable dictionary, an object tested for belonging to the class vegetation canopy removal or growth must exhibit change index values and texture values, respectively, slightly higher and slightly lower with respect to those requested during the reliable dictionary loop. This enforcement of the requirements for accepted objects is useful to avoid uncontrolled growth of detection patterns and to increase the overall robustness of the system.

It is worthwhile to remark that this methodology is able to detect cyclical changes, provided that, as depicted in Fig. 1, the state of the specific terrain between t_0 and t_2 is different. This is because the information layers used for the detection are essentially bi-temporal. The fully multitemporal information is exploited to improve the despeckling quality

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(a) (b)

Fig. 9. SAR collage. (a) Multitemporal RGB composite. (b) Semantic edge map.

and the overall calibration accuracy of the time series [51]. Obviously, a higher texture characterization, as a consequence of deep speckle removal, is beneficial for the effectiveness of the textural information layer.

C. Modularity

One of the characteristics of the proposed architecture is its modularity which allows to increase the semantics and prior knowledge in input to the system. This is useful to cope with nonstandard situations described in Section II-A.

As an example, the collage depicted in Fig. 2(b) is considered. Its multitemporal color composite version, produced according to [13], is depicted in Fig. 9(a). It is possible to observe that layover patterns in the third quadrant counterclockwise exhibit a greenish color which testifies a higher backscattering in the test image with respect to the reference one. However, this is due to the instability of the SAR signal over highly reflective surfaces rather than growth patterns [70]. Nevertheless, these layover regions are likely to be misclassified as growth using standard literature change detection methods.

To avoid this, it is possible to consider the texture evidence in the system according to the diagram depicted in Fig. 10(a). Its difference with respect to the workflow previously introduced is the presence of a new processing branch dedicated to the semantic texture extraction.

In this case, the dictionary relevant to vegetation canopy removal or growth patterns is supported by a second dictionary collecting color labels identifying edges. Given the composition of the input product, it contains blue tonalities. The obtained edge map [see Fig. 9(b)] is processed for the extraction of a simple texture measure, that is, the Nagao–Matsuyama texture [71]. It is a box counter. Given a sliding window of size w (reliable dimensions are in the order of 70–90 m), the central pixel is classified as high-texture area if the number of edge points falling within the calculation window is $n > 2w+ 1$. Otherwise, the point is classified as belonging to a low-texture area.

The new texture map contributes to the decision phase, which now requires accepted objects to lay into a low-texture area. Considering the edge map depicted in Fig. 9(b), it looks evident that being layover and sand areas characterized by unstable backscattering and high texture, this workflow allows for masking out false alarms without significantly penalizing the detection performance. This will be quantitatively assessed in Section III.

Another critical situation that can be addressed exploiting the modularity of the proposed architecture is that of two changes causing a backscattering with similar characteristics, like flooding in a dynamic agricultural area. This is the case of the Albufera scene [see Fig. 2(c)], in which the objective is to discriminate vegetation canopy removal from flooded areas. To this end, the workflow reported in Fig. 10(b) is suggested. In this workflow, vegetation canopy removal and flood are analyzed starting from different dictionaries. Because of the similarity of the two phenomena, these dictionaries partially overlap (especially in their unreliable part) and generate ambiguity areas. These areas are processed with a dedicated OBIA aiming at solving area conflicts between the two classes. At the end of the second defuzzification, only regions reclassified as belonging to the vegetation canopy removal class are kept in the final map.

The importance of modularity arises when dealing with deforestation applications. Deforestation and harvesting are two subclasses of the vegetation canopy removal class having the same characteristics, both being characterized, at microwaves wavelength, by a drop in the backscattering. However, in SAR images, deforestation is accompanied by another significant phenomenon, related to texture.

Let us consider the situation depicted in Fig. 11(a). This close-up is taken from a multitemporal RGB composite of the Brazil data set built using two images acquired with a six-day time span, to represent a situation of undisturbed landscape.

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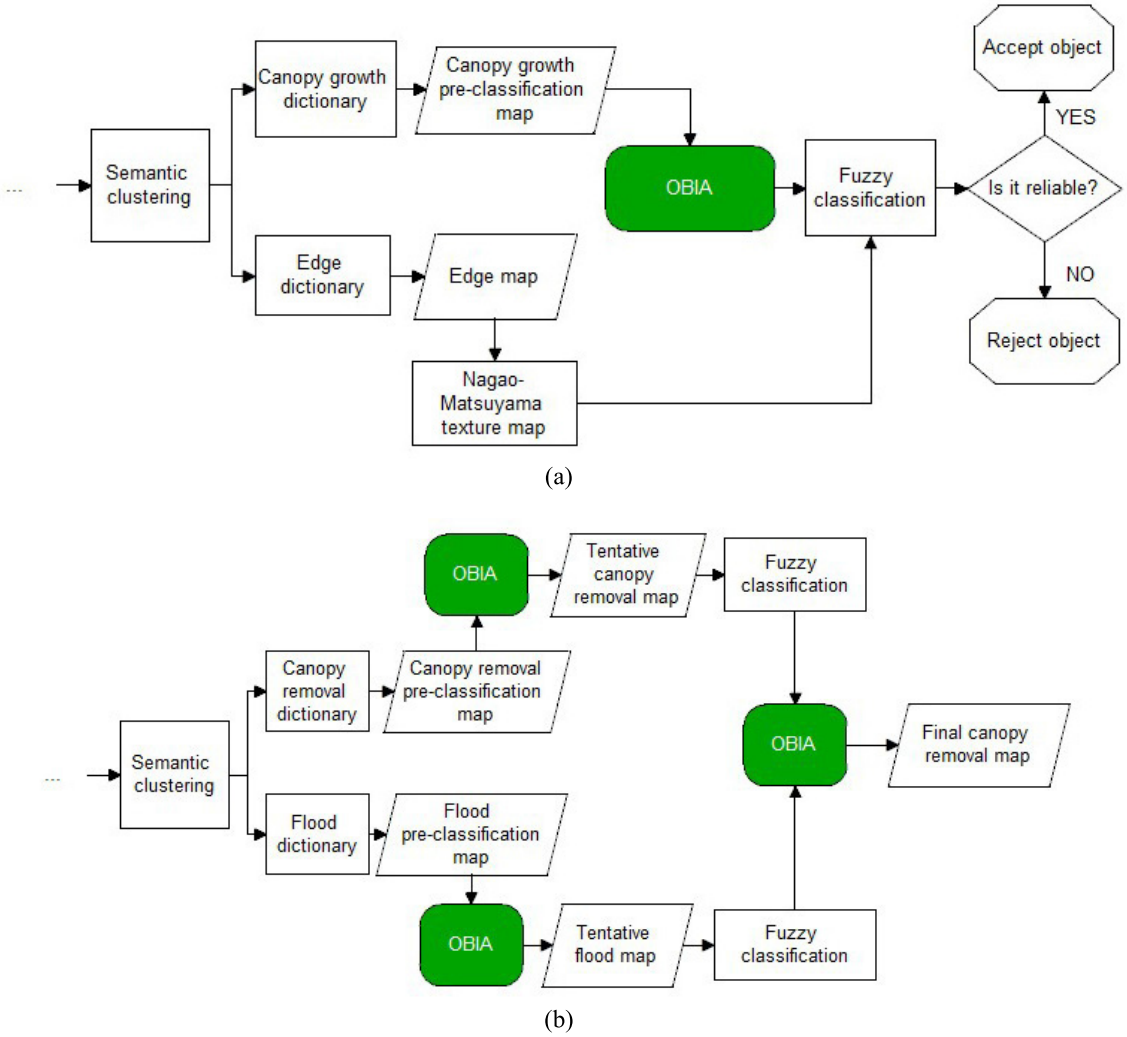


Fig. 10. Workflows for coping with (a) high texture features and (b) multiple changes in the study area.

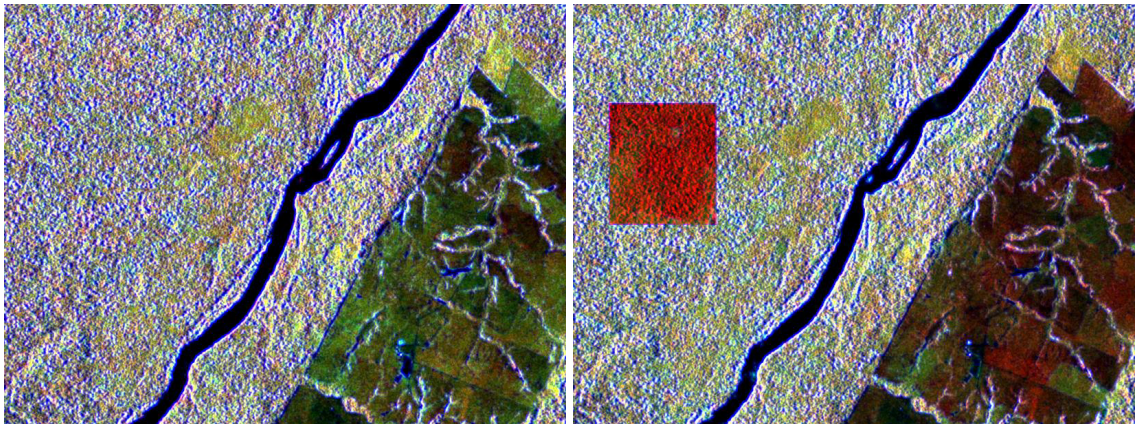


Fig. 11. Effects of clear-cut deforestation patterns on multitemporal RGB SAR images. (a) Close-up taken from a product obtained using two images acquired with short revisit time, i.e., with undisturbed forest. (b) Particular of the simulated deforestation pattern.

It is clear that the forest is rendered by a pattern of pure texture. The same area, after the introduction of the simulated deforestation pattern, is depicted in Fig. 11(b). It introduces two major effects, the first on the backscattering, and this is visible from the dominance of the vegetation canopy scattering in the reference image (red channel). The second one, even more significant, is on the texture, which is completely destroyed. This suggests that when dealing with deforestation, the texture layer to be used as input for the fuzzy classifier is not the data range texture calculated on the product reported

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in Fig. 11(b) [see (1)], but the texture change index between the two representations calculated as follows:

$$\mathrm{TCI} = \frac{d_{r_{t_1}} - d_{r_{t_2}}}{d_{r_{t_1}} + d_{r_{t_2}}}, \quad \mathrm{TCI} \in [-1, 1] \tag{4}$$

where $d_{r_{t_1}}$ and $d_{r_{t_2}}$ are the data range calculated [according to (1)] on the multitemporal RGB product obtained from images acquired with short revisit time and the data range calculated on the multitemporal RGB product in which the test image is the one closing the observation period.

The introduction of this new texture layer is helpful in the discrimination of deforestation because the drop in the texture due to clear-cut is significantly more informative than the data range measure which, as discussed in Section II-B, is calculated starting from the test image only. Therefore, the information it can carry is simply the presence of a bare soil, which is the same situation determined by harvesting.

Thus, with respect to the workflow reported in Fig. 3, the identification of deforestation patterns involves two RGB composites sharing the same reference image (acquired at the beginning of the observation period) and with test images representing one the situation of undisturbed landscape and the other of the observation period. The benefits brought by this workflow are quantitatively assessed in the relevant discussion provided in Section III.

D. Reference Literature

The literature exploited as benchmark for the proposed methodology is briefly recalled in the following subsections for the ease of the reader. In particular, the log-ratio (LR) [28], the cumulative change index (CCI) [35], the absolute change estimator (ACE) [72], and the generalized maximum likelihood ratio test (GLRT) [32] are summarized here. All of them are pixel-based techniques requiring thresholding for the extraction of relevant changing image segments. In the following formulas, the images are assumed to be in the intensity format.

1) Log Ratio: The log ratio (LR) is simply defined as the ratio between the pre and postevent images expressed in logarithmic scale [28] as follows:

$$\mathrm{LR} = \log \frac{I_{t1}}{I_{t2}} = \log I_{t1} - \log I_{t2} \tag{5}$$

where I_{t1} and I_{t2} represent the pre and postevent images, respectively.

2) Cumulative Change Index: This change indicator has been defined by [35] for a bitemporal change detection problem like the mapping of temporary water surfaces in the semi-arid environment (which is equivalent to flood mapping). Here, the multitemporal version of that indicator, the CCI, is exploited as benchmark. It is defined as follows:

$$\mathrm{CCI} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1, j>i}^{N} \frac{I_i - I_j}{I_i + I_j} \tag{6}$$

where N is the number of images in the considered time series, and i and j, with $j > i$, represent two of its generic elements.

3) Absolute Change Estimator: The ACE algorithm has been developed in [73]. As explained in [72], the method exploits backscattering values and their temporal changes by computing the logarithm of the ratio between any date within the multitemporal data set, composed of N images, as follows:

$$\mathrm{ACE} = 10 \ln \left[\frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{j>1}^{N} R_{ij} \right] \tag{7}$$

where R_{ij} is the maximum of the intensity ratios (calculated pixel-wise) between the generic images i and j of the time series, that is

$$R_{ij} = \max \left(\frac{I_i}{I_j}, \frac{I_j}{I_i} \right) \tag{8}$$

The map of changes is finally obtained by segmentation of the ACE map.

4) Generalized Maximum Likelihood Test: The GLRT has been introduced in [32] and is defined as follows:

$$\Delta = \frac{\sqrt[N]{\prod_{i=1}^{N} \eta_i}}{\frac{1}{N} \sum_{i=1}^{N} \eta_i} < \lambda \tag{9}$$

where $\eta_i = \sum_{i=1}^{w} z_{w,i}$ is the total intensity of the generic image segment composed by w pixels with reflectivity z. In other words, the GLRT is defined by the ratio between the (temporal) arithmetic and the geometric means of the intensity of the image segment.

III. Experimental Results

In Fig. 12, some patches extracted from the available data relevant to vegetation canopy removal and growth patterns are shown. In particular, the first row of the picture depicts the reference data, while the second the corresponding area as rendered on the multitemporal RGB products exploited as input for the workflow. This picture is also useful to understand the differences between the multitemporal SAR representation and NDVI changes. As an example, looking at the Cambodia case study in Fig. 12, it is possible to appreciate the nonperfect correspondence between the red patterns in the SAR picture [see Fig. 12(c)] and the areas characterized by an NDVI drop of 0.3 [see Fig. 12(g)]. This means that the change detection success is also related to the SAR sensitivity to the amount of biomass change and that areas presenting comparable changes at microwaves actually exhibit higher discrepancy in the MS domain which can be source of misclassifications. This topic will be quantitatively addressed in Section III-A.

The experimental results for the five analyzed data sets are summarized in Figs. 13 and 14, which report the receiver operative curves (ROCs) for the proposed methodology and for the literature methods used as benchmark. In particular, Fig. 13 refers to the Vietnam and Cambodia data sets, while Fig. 14 to the Collage, Albufera, and Brazil ones.

For ACE, CCI, LR, and GLRT methods, the ROC is built by recursively applying a different threshold value (with a fixed step of 0.1 between one value and the following) to the relevant change indicator. The ROC for the proposed methodology is

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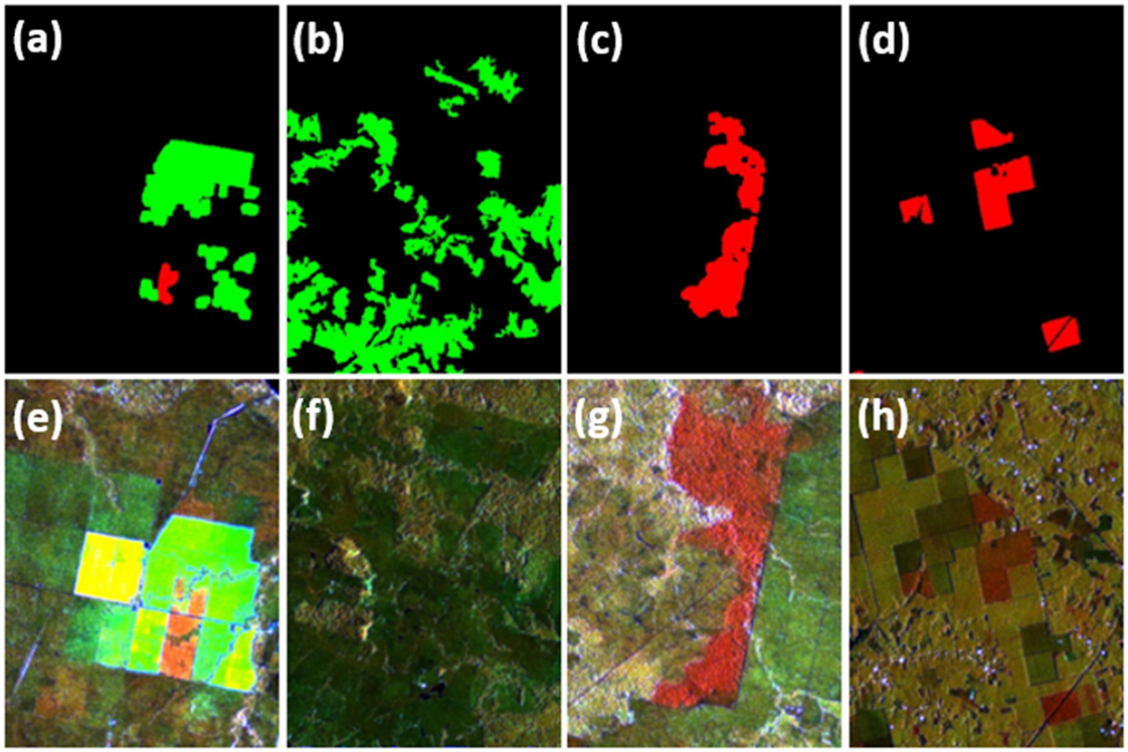


Fig. 12. Some vegetation growth and removal patterns extracted from the available reference data and multitemporal RGB images. (First row) Reference data for (a) Cambodia—canopy growth, (b) Brazil—canopy growth, (c) Cambodia—canopy removal, and (d) Vietnam—canopy removal. Green and red clusters represent vegetation growth and removal, respectively. (Second row) RGB patches for (e) Cambodia—canopy growth, (f) Brazil—canopy growth, (g) Cambodia—canopy removal, and (h) Vietnam—canopy removal.

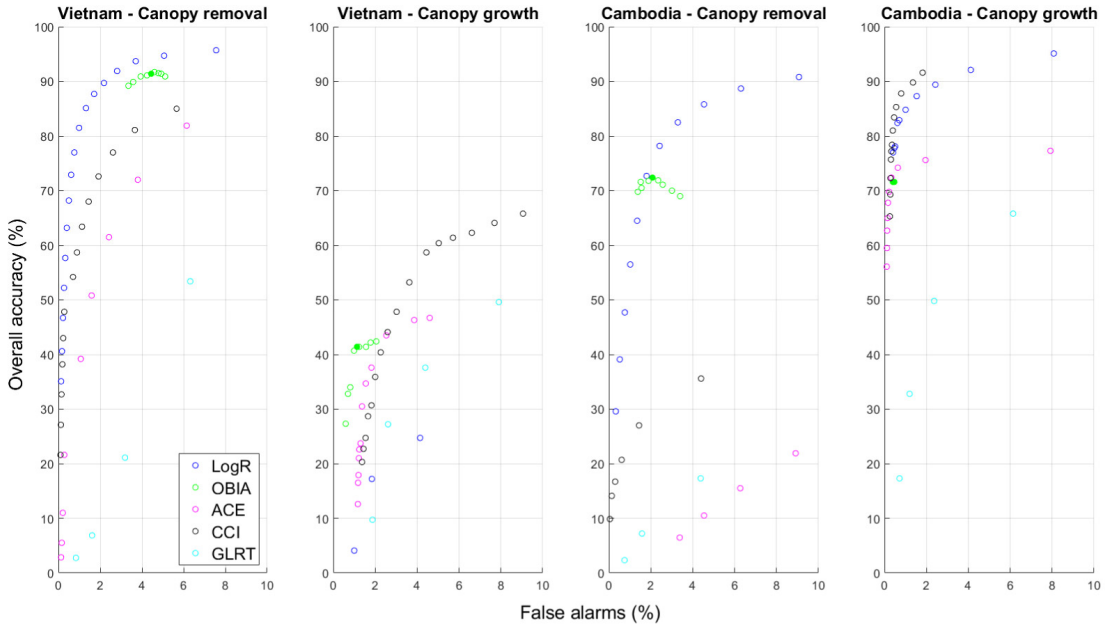


Fig. 13. ROCs for the Vietnam and Cambodia data sets. Green dots refer to the proposed method (OBIA). The solid marker corresponds to default parameters' setup in the fuzzy classification system. The color code for the other methods is blue dots—LR, red dots—ACE, black dots—CCI, and cyan dots—GLRT.

obtained by increasing and decreasing the fuzzy classification parameters with respect to the default setup of ±5%, ±10%, ±15%, and ±20%. In general, a satisfying ROC should occupy the upper left portion of the plane defined by the percentage of false alarms (considered in the graphics in the interval [0, 10%] to improve the readability of the plots) and the percentage of correct decisions. The color code for the displayed ROC is blue dots—LR, red dots—ACE, black dots—CCI, cyan dots—GLRT, and green dots—OBIA. The solid marker in the OBIA ROC corresponds to default fuzzy parameters.

In the following, the different panels of Figs. 13 and 14 will be referred starting from the left one.

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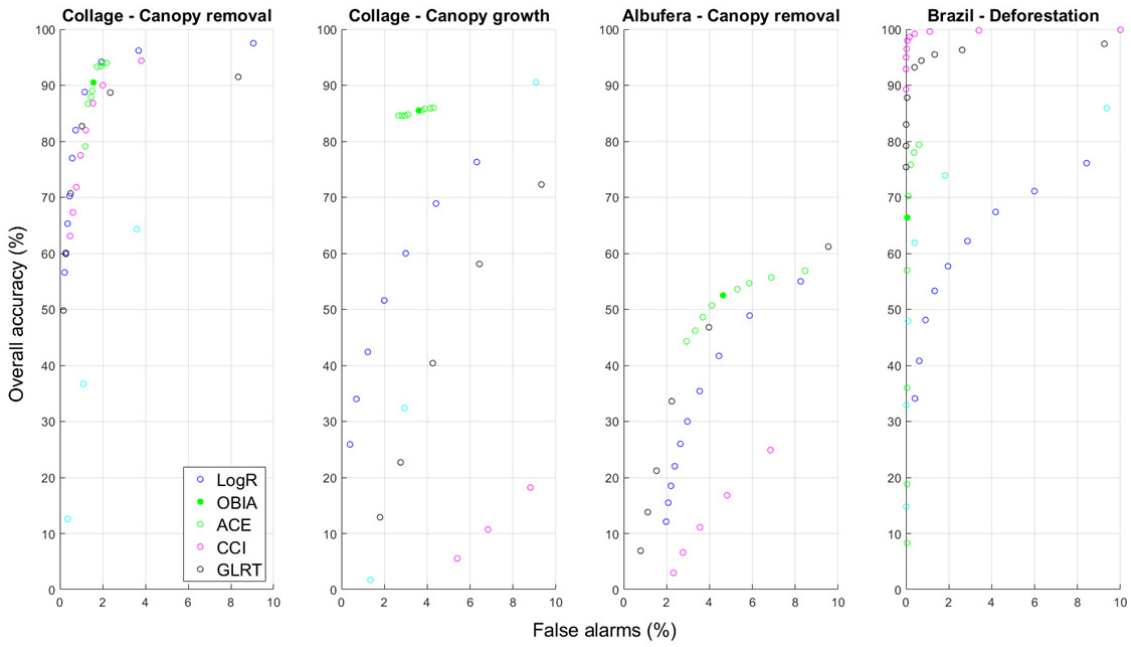


Fig. 14. ROCs for Collage, Albufera and Brazil data sets. Green dots refer to the proposed method (OBIA). The solid marker corresponds to default parameters' setup in the fuzzy classification system. The color code for the other methods is blue dots—LR, red dots—ACE, black dots—CCI, and cyan dots—GLRT.

The first experiment, reported in the first panel of Fig. 13, refers to canopy removal patterns detection for the Vietnam data set. In this case, the best ROC is that of LR, which is very steep and shows, on average, high detection rates with a low percentage of false alarms. The ROC of the proposed methodology is placed just below it and is insensitive to the variations in the fuzzy parameters being almost flat. Fixing the false alarms, the CCI shows a detection rate about 10% lower than the aforementioned methods. The ACE shows even worse performance, while the GLRT is unsatisfying, its ROC being almost placed on the plane bisector. The best results obtained by the methods using impulsive information layers can be explained by the fact that harvesting is an impulsive phenomenon as well; therefore, the detection rate of cumulative methods is a function of the time of the event. The closer the end of the observation period, the more dominant the previous conditions of stability (or even growth) of the vegetation cover.

In the second experiment, whose results are reported in the second panel of Fig. 13, the Vietnam data set has been tested for vegetation canopy growth. Unlike harvesting or deforestation, this phenomenon is not impulsive, but develops over time with increasing intensity. Therefore, in this case, cumulative information layers, like the CCI, the ACE, and the GLRT, tend to perform better than the LR, which has a very low detection rate and, in general, an ROC with moderate slope. Moreover, growth patterns are not very well-defined on the SAR image like in the reference MS and this further hampers the detection of the relevant regions, making the detection rate for all the experiments rather low. This also applies to the OBIA experiment. The characteristics of the scene make a limited number of regions falling in the core reliable dictionary, with unreliable dictionary clusters succeeding in aggregating pixels up to a median detection rate of about 40% with a false alarm rate around 1%. Unlike the previous experiment, the OBIA ROC is now more sensitive to the parameters of the fuzzy classification system, the performance being falling down as the requirements for accepted objects are strengthened. This is because most of the detection is due to clusters belonging to the unreliable dictionary, and so without the characteristics of change index and texture expected for growth patterns.

The results obtained in the Cambodia experiment follow the trend of those just discussed. As for vegetation canopy removal, see the third panel of Fig. 13, the LR is the best performing, showing an ROC quite steep in the left portion of the plane. The OBIA ROC is, again, very flat. Almost all the experiments have an accuracy higher than 70% and a false alarm rate ranging from 1.38% and 3.4%. All the other methods show poor performance, with particular reference to the ACE.

In the Cambodia vegetation canopy growth experiment, whose results are reported in the last panel of Fig. 13, the methods based on cumulative information layers, especially the CCI and the ACE, show a significant improvement in the performance. The LR also has a steep ROC, almost overlapping with that of the CCI. In this case, the OBIA ROC is in this case a single point, with all the experiments having an accuracy of about 70% and a false alarm rate lower than 1%. This means that the detection is mainly due to very well-defined growth patterns belonging to the reliable dictionary. Conversely, the unreliable dictionary is dominated by land features, which are discarded by the fuzzy classification system when aggregated to real growth patterns identified by the reliable one.

The first panel in Fig. 14 reports the results of the experiment relevant to vegetation canopy removal in the

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collage scene. As discussed before, ambiguities in this scene are more related to canopy growth. Therefore, all the ROCs, except that concerning the GLRT, are very steep, almost overlapping, and located in the low false alarm rate area of the plane.

The situation is different if the second panel of Fig. 14, showing the results for the vegetation canopy growth detection experiment, is considered. In this case, the ambiguities due to the presence of layover and sand affects the performance of all the pixel-based methods, which show smooth ROCs located in the lower part of the plane. Considering the same false alarm rate, the OBIA increases the detection performance of a percentage ranging between 20% and 25% with respect to the best performing literature method, which is the LR in this case. Moreover, even in a critical situation, the stability of the method with respect to the fuzzy classification parameters is confirmed, the ROC being almost flat. In this experiment, the poor performance of the ACE detector is remarkable, whose ROC witnesses a completely failed detection.

The third panel in Fig. 14 shows the results of the vegetation canopy removal experiment implemented over the Albufera natural reserve. As previously discussed, this is a nonstandard scene because the area is periodically flooded due to the presence of rice cultivations. This means that it is possible, like in the situation here considered, that two changes with similar effects on the backscattering occur in the study area. The obtained results show that the proposed OBIA is the best performing among the considered techniques, especially for false alarm rates ranging between 2% and 4%. In this interval, the proposed method presents an accuracy of about 5% higher than the CCI and about 10% higher than the LR. The ACE has a very smooth and poorly peaked ROC, and thus the results are characterized by low detection and high false alarm rate. As for the GLRT, there are no ROC points in the considered false alarm rate interval, and therefore, the detection fails using this method.

The last panel in Fig. 14 shows the results concerning the Brazilian Amazon deforestation experiment. As previously discussed, deforestation is a subclass of the vegetation canopy removal class. Therefore, the objective is to isolate clear-cut patterns from agricultural harvesting ones. In this application, as discussed in the previous section, the texture drop due to deforestation is very informative and helpful in discriminating the relevant patterns from agricultural ones. This makes the proposed OBIA very effective, the accuracy of the experiment being, run with default fuzzy parameters, in the order of 65% with a false alarm rate of 0.06%. This means that the proposed methodology is able to detect deforestation patterns excluding all the huge harvesting patterns present in the scene. This is not the case for LR, which shows a very smooth ROC. The drop in the classification accuracy is in the order of 30% considering a fixed probability of false alarm.

In this application, the ROC of the proposed OBIA is very steep. The slackening of the parameters, especially the textural one, necessary to have deforestation patterns accepted by the fuzzy classifier, makes the detection rate rapidly increase up to almost 80%, with an increase in the false alarm rate rather small and up to 0.64%. Conversely, the enforcement of the parameters makes the detection accuracy fall down very quickly to zero, with unchanged false alarm rate with respect to the default experiment. This is because the texture requirements in the default setup are very high in the attempt of trying to exploit the informative content of the layer in minimizing false alarms. This means that the enforcement of the parameters is useless and counterproductive because there is a risk that they become out of scale.

As for methods using cumulative information layers, the results are very satisfying, especially those of CCI and ACE, which show the steepest ROCs. However, this is mainly due to the positioning of the change within the time series. As previously discussed, CCI and ACE tend to work well when the change is placed in the initial part of the time series, with detection performance decreasing as the change shifts toward the end of the observation period. The change being, in this case, placed at the middle of it, these two methods, with GLRT at lesser degree, ensure almost perfect detection with low false alarm rate.

In Table III, the aggregated data extracted from the ROCs discussed above are reported. In particular, they refer to an ideal optimum operational regime of false alarms ranging between 0.5% and 5%. In the table, OA stands for overall accuracy (percentage), FA for false alarm rate, and FO for number of false objects with an extension greater than 50 000 m^2.

As for vegetation canopy removal detection experiments, from the table it can be seen that the proposed OBIA outperforms the reference literature in three experiments out of five concerning the detection accuracy, with an average value of 74% across all the tests. The average registered false alarm rate is 2.45%. In this particular indicator, the best performance is that of CCI (1.18%), but with a detection accuracy significantly lower (on average about 26%) compared with the proposed methodology.

The LR does not show particular peaks in the performance and tend to have average values for all the performance indicators, except the detection accuracy for the Albufera experiment, which is quite low. The ACE method is the best performing in the deforestation detection experiment (as discussed before, this is due to the positioning of the clear-cut within the time series) but failed in the Cambodia and Albufera test cases. The GLRT is the worst performing among the considered methods, showing one experiment totally failed (no ROC points in the considered false alarm rate interval available) and detection values quite low across all the tests.

One interesting parameter to be evaluated is the object-based false alarm. In this case, the proposed OBIA is the best performing even if the percentage of false alarms (evaluated pixel-based) is not the lowest one. This means that this method tends to overestimate real vegetation canopy removal patterns through the aggregation of stable land clusters, mainly belonging to unreliable dictionary, to the detection given by reliable dictionary clusters. Conversely, the LR, showing a false alarm rate lower than 0.55%, detects a more than doubled number of false objects, therefore tending to disrupt detection patterns. A similar behavior, even if at lower degree, is shown by the ACE and the GLRT. The CCI, which has the smallest

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TABLE III

SUMMARY OF THE PERFORMANCE OF THE ANALYZED METHODS EXTRACTED FROM THE ROCS REPORTED IN FIG. 13. DATA REFER TO AN IDEAL OPTIMUM FALSE ALARM RATE INTERVAL RANGING BETWEEN 0.5% AND 5%. OA—OVERALL ACCURACY (%), FA—FALSE ALARM RATE (%), FO—NUMBER OF FALSE OBJECTS WITH EXTENSION HIGHER THAN 50 000 m^2. BOLD CHARACTERS IDENTIFY THE BEST REGISTERED PERFORMANCE FOR THE SPECIFIC INDICATOR

Vegetation canopy removal															
	ACE			CCI			LR			GLRT			OBIA		
	OA	FA	FO	OA	FA	FO	OA	FA	FO	OA	FA	FO	OA	FA	FO
Vietnam	56.1	2.00	90	70.3	**1.68**	**77**	89.7	2.18	166	14.0	2.40	337	**91.1**	4.44	194
Cambodia	10.5	4.54	979	27.0	**1.44**	**361**	**74.4**	2.11	386	7.21	1.58	505	71.1	2.08	383
Collage	79.8	1.09	143	82.7	**1.03**	180	85.4	0.95	**134**	36.7	1.10	142	**90.5**	1.57	151
Albufera	8.84	3.15	144	21.2	**1.53**	111	26.0	2.64	154	na	na	na	**50.7**	4.11	**59**
Brazil	**98.0**	0.07	46	90.5	0.22	117	55.6	1.65	904	47.9	0.10	72	66.4	**0.06**	**11**
Average	50.6	2.17	280	58.3	**1.18**	169	66.2	1.90	349	26.4	1.29	264	**74.0**	2.45	**160**
Vegetation canopy growth															
	ACE			CCI			LR			GLRT			OBIA		
Vietnam	23.1	1.27	27	40.4	2.26	113	17.2	1.84	95	27.7	2.61	148	**41.0**	**1.13**	**66**
Cambodia	74.2	0.64	925	86.5	0.68	350	**82.9**	0.71	242	32.8	1.20	1294	71.6	**0.43**	**191**
Collage	na	na	na	22.7	2.76	241	47.0	**1.61**	246	17.0	2.14	**158**	**85.5**	3.60	250
Average	48.6	**0.95**	489	49.9	1.90	234	49.0	1.38	194	25.8	1.98	533	**66.0**	1.72	**169**

average registered false alarm rate, is quite consistent with respect to the number of false objects, which is slightly higher than that detected by the proposed OBIA.

This parameter assumes particular relevance in deforestation applications. In the relevant experiment, the proposed OBIA showed to be extremely robust to the indicator with a median value of 11 false objects detected. All the other methods present a much higher number of false objects detected, ranging from 46 of the ACE to 904 of the LR.

As for vegetation canopy growth experiments (see the lower part of Table III), the obtained results mainly confirm what stated above. The proposed OBIA is again the best performing concerning the total average detection accuracy and the average number of false objects detected. The best detected average false alarm rate is given by the ACE, even if there is one experiment out of three (the SAR collage one) which completely failed. The LR has outstanding performance only in the Cambodia experiment (best detection rate registered) but starts to fail in critical situations like the Vietnam scene and the SAR collage and, on average, shows a detection rate of 17% lower compared with the proposed OBIA. The CCI almost failed in the SAR collage experiment, while the GLRT was again the worst performing, registering the lower performance indicators among the considered literature.

Summarizing, the proposed OBIA represents a robust solution to change detection of natural land cover registering the best performance in almost all the considered classification quality indicators, namely, the detection rate and the number of false object detections. Its robustness has been quantitatively assessed, with no occurrence of failed classifications, in both standard and critical situations, that is, those characterized by the presence of typically unstable targets and/or multiple changes in the study area. Another indicator of the stability of the proposed architecture is given by the scarce sensitivity to parametrization, as suggested by the functioning points (i.e., ROC points) on average closer to each other compared with the typical monotonic trend of thresholding segmentation.

Among the analyzed pixel-based reference literature, the LR is probably the most reliable choice, with limited occurrence of failed classifications (namely, the Vietnam vegetation canopy growth, the Albufera vegetation canopy removal one, and the Brazilian deforestation ones) and a quite high detection accuracy. Moreover, it represents a very quick and easy solution to implement. However, as for all pixel-based methods, the weakness related to thresholding holds, making it not suitable for operational unsupervised scenarios.

A. Sensitivity to the Amount of Biomass Changes

The ROCs discussed in the previous paragraph represent the picture for biomass changes equivalent to a change in the NDVI equal to ± 0.3. The study of the sensitivity of the methodology with respect to this parameter is now in order.

In Fig. 15, the ROCs for the different examined methodologies as a function of the NDVI difference threshold are depicted. The reference experiment is that of canopy removal within the Vietnam scene. The threshold values considered to extract the reference data for changes are $[-0.1, -0.2, -0.3, -0.4, -0.5]$. Compared with the value of -0.3 used before, a decrease to -0.4 or -0.5 is equivalent to be considered as changed areas interested by an increased removal of biomass. Conversely, an increase in the threshold to -0.2 or -0.1 means to be considered as changed areas interested by a smaller removal of biomass. Finally, it is worthwhile to note that with respect to the performance assessment presented before, the worst performing information layer, that is, the GLRT, has been substituted, as suggested in [37], by a thresholding applied on the KLD layer calculated using the first and the last images of the time series as input. Useful libraries for this purpose have been found in [74].

The first panel from the left of Fig. 15 is relevant to the proposed OBIA. From the ROCs, it arises that the method maintains its insensitivity to its parameters (i.e., the ROCs are quite flat) and that is much more sensitive to an increase in the threshold than to its decrease. The same trend is mostly observed for all the other considered methodologies, with the exception of LogR, which is quite insensitive to decrease in the thresholds. This means the removal of biomass corresponding to an NDVI difference of -0.3 has saturated the detection capacity of the technique since no significant variations are

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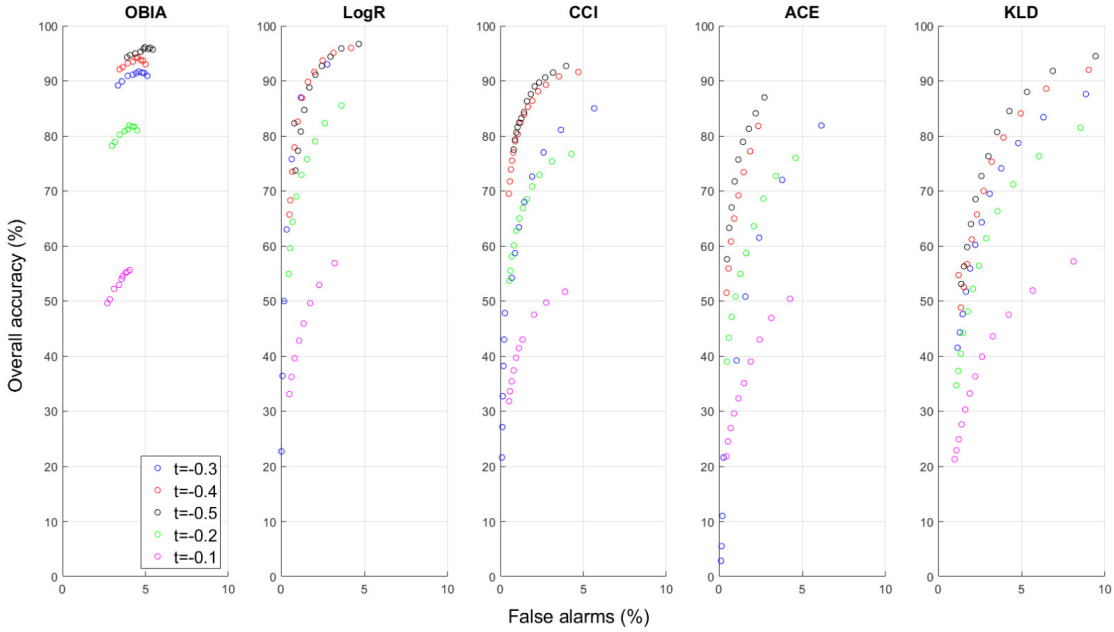


Fig. 15. ROCs for the different considered methods as a function of the variation in the threshold applied to the NDVI difference map to retrieve reference data. (Left to right) First panel—proposed OBIA, second panel—LogR, third panel—CCI, fourth panel—ACE, and fifth panel—KLD.

observed when the minimum level of biomass removal to be detected is increased to NDVI differences of -0.4 or -0.5. The newly added KLD thresholding had performance comparable with those obtained using the other methodologies used for benchmarking.

All the methods show poor performance when the minimum amount of biomass removal to be detected is set to an NDVI difference of -0.1. However, in this case, it is possible that the reference data are significantly overestimated. Such a small threshold on the NDVI difference can bring to the inclusion within the reference map of patterns not associated with the variations in biomass, also due to a nonperfect compensation of the atmosphere. As a general comment, based on this analysis it is possible to argue that the more the biomass is removed, that is, the more abrupt the change, the best the performance of the techniques. This is consistent with the complex SAR backscattering mechanism from vegetation canopies which tend to make more evident abrupt changes like harvesting or deforestation [75]. Moreover, the general stability of the false alarm rate, irrespectively of the minimum change in the biomass to be detected, is an indicator that part of the changes detected by using SAR are related to different phenomena. Among them, the variations in humidity have been variously reported in the literature to be responsible of the alterations of the backscattering from vegetation canopies [76].

B. Role of the Dictionary and of the Information Layers

In the proposed methodology, the dictionary is used to create a preclassification mask which, being based on the verbal attributes most likely to be associated with the phenomenon under study, favors the exclusion of areas characterized by similar temporal backscattering characteristics but substantial difference in the semantics of changes. As an example, wind-induced roughness variations on water surfaces or variations in the moisture content in a bare soil can cause changes in the reflectivity function of the scene similar to those due to vegetation canopy removal or growth phenomena. The color attributes included in the dictionary related to the specific phenomenon, condensing backscattering and textural characteristics, constitute a first tool useful to mitigate false alarms. However, the preclassification mask is in any case likely to be overestimated [53] and needs further refinement through fuzzy analysis of the identified clusters.

The choice of the words belonging to the reliable and unreliable dictionary is empirical and clearly affects the change detection performance. As suggested in [53], this operation can be implemented by considering patch visually identified as a changing pattern and reporting the attributes of the clusters falling within it, subdividing them in the reliable and unreliable dictionary based on their occurrence in the test area. This operation could be time-consuming, depending on the expertise of the operator, but is part of the preliminary training of the learning algorithm and should be done just once thanks to the stability of the color rendering of the input RGB products [51].

The contribution of the different parts of the dictionary to the final detection is different. Considering the Vietnam canopy removal experiment as reference, the reliable dictionary and the first element of the unreliable one are responsible for about 60% of the total detections. Computational time varies as a function of the number of images contained in the time series and the size of the considered study area. To give an idea, the Vietnam experiment (58 images covering an area of 1800 km^2) has been concluded (excluding the preprocessing phase) in about 1 h, with the total time subdivided in about 10 min for the generation of the RGB minutes for clustering and 10 min to classify.

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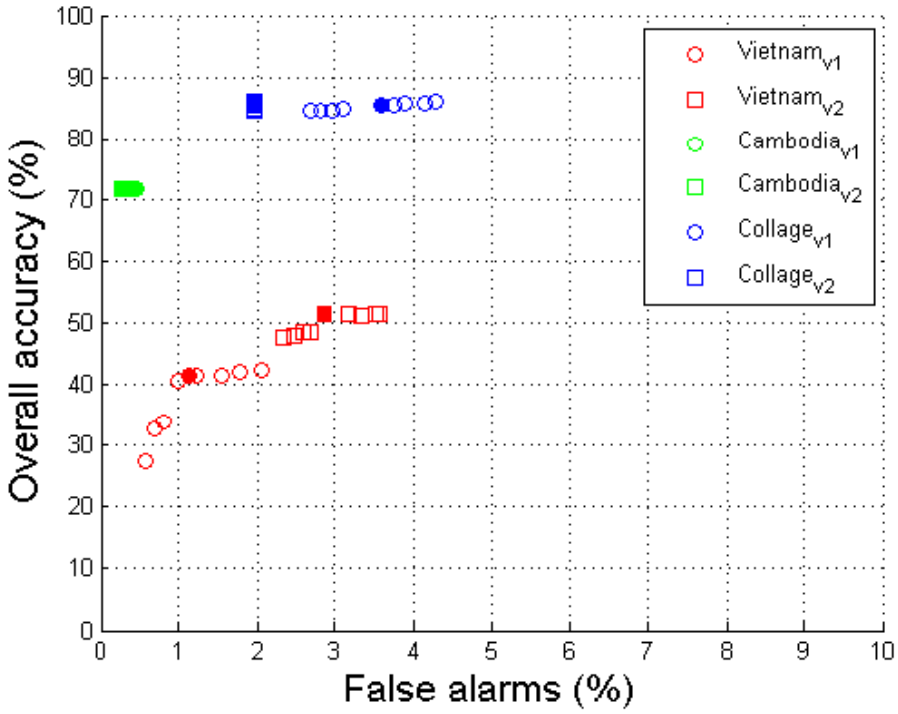


Fig. 16. Comparison of the ROCs obtained feeding the fuzzy system with the bitemporal (circular markers) or the multitemporal (square markers) scattering layer. Vietnam—red curve, Cambodia—green curve, and Collage—blue curve.

Following the approach proposed in [71] for optical aerial pictures, the methodology can be generalized acting on both the dictionary and the information layers. Different words can be used to preclassify different image features having different geometric, scattering, and textural properties [77]. As an example, lakes and rivers, both belonging to the macroclass of water bodies and therefore sharing the same dictionary, can be identified and separated exploiting their different geometric characteristics leading, in the first case, to high compactness [53], in the second to significant elongatedness [77]. This means that further research is needed to assess the best combination of information layers to classify the changes in vegetation canopies and to explore the possibility of the joint exploitation of more texture and/or geometric features toward deep architectures. As an example, the discrimination between flooded areas and pattern of removed vegetation has been here implemented by exploiting the shape of the relevant fuzzy set used to classify. However, it is possible that the introduction of further information layers taking into account the peculiar texture and geometric pattern of floods and agricultural fields can be beneficial in the discrimination of phenomena exhibiting similar backscattering behavior.

C. Strategies for the Enforcement of the Classification System

One of the outcomes of the analysis of the experimental results is that the exploitation of cumulative change layers, like the CCI or the ACE, tends to provide better performance compared with the usage of bitemporal ones. The high flexibility of the proposed methodology allows for easily exploiting this finding to enforce the classification system.

The principal problem of the usage of the CCI or the ACE within our fuzzy classifier is the definition of the fuzzy set parameters, as the values of these quantities depend on both the magnitude of changes and the number of images. However, the availability of a dictionary related to the specific phenomenon (i.e., the growth of vegetation) is helpful for overcoming this obstacle as its statistics can be used to opportunely calibrate the fuzzy set parameters.

To this end, the masks representing the reliable and unreliable parts of the dictionary are extracted and applied, in our case, to the CCI layer. The mean and the standard deviations of the pixels belonging to them are used to set up the fuzzy set parameters, establishing a direct link between the dictionary and the classifier. Then, the system is run as previously described.

The results of the experiments implemented with this modification are reported in Fig. 16. In this graph, the results of the system using the bi-temporal change layer are depicted using the circular marker. Those obtained using the multitemporal change layer are represented by the square marker. It arises that in the case of the Vietnam experiment (red markers), a significant improvement in the detection rate (of about 10%) is obtained by exploiting the cumulative change layer, with a negligible increase in false alarms. As for the Cambodia experiment (green markers), the usage of the cumulative or the bitemporal change layer was mostly equivalent. Finally, in the case of the collage (blue markers), a slightly lower than 5% increment of the average detection accuracy and a decrease of about 1.5% of the false alarm rate have been registered. Moreover, a very stable ROC, basically a point, has been restituted by the experiments.

IV. Conclusion

In this work, a new architecture exploiting the SAR data for the detection and labeling of changes in natural land cover has

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been introduced. The proposed methodology takes advantages of consolidated SAR preprocessing practices, introducing the main novelties in the information extraction process, which benefits new input products and concepts relating traditional SAR remote sensing with modern GEOBIA. Unlike the past literature, in which the change detection is mostly implemented through thresholding, the proposed methodology aims at the exploitation of the intrinsic semantics of data to allow for the identification of the classes of interest, avoiding confusion with phenomena having similar characteristics when observed at microwave wavelengths.

The proposed methodology has been tested on five different data sets with experiments involving agriculture and forestry applications and the obtained results were compared with those given by the reference SAR change detection literature. They indicate that in most of the cases, the LR segmentation is able to enhance the presence of changing patterns, provided that reliable thresholding is implemented. The usage of more complicated change indicators could be counterproductive, since the complexity increment they introduce is often not justified by a significant performance enhancement.

The most critic drawback using SAR in change detection applications is the false alarm rate, which is directly related to the reliability of thresholding procedures and to their impossibility in labeling changes. The proposed architecture introduces novelties to overcome these limitations. First, the exploitation of semantic clustering allows for directing the processing toward the identification of a precise class of changes through the usage of a class-specific dictionary. In the decision phase, fuzzy logic principles allow for the fully unsupervised combination of opportune information layers, thus avoiding thresholding for the extraction of changing patterns. Quantitatively, compared with the best performing literature method, the use of the proposed methodology introduced an increase in the detection accuracy of about 8% and 17% in vegetation canopies removal and growth experiments with comparable false alarm rates. This results in an enhancement of the overall robustness of the detection system, which represents a step forward toward the full exploitation of SAR images in continuously monitoring operative scenarios.

The proposed methodology can be generalized for the extraction of different image features through appropriate variations in the dictionaries and of the object properties used to classify, even if further research is needed to assess the effectiveness of the exploited information layers and the possibility to jointly exploit more texture and geometric features configuring the setup of a deep architecture. Further investigations are needed, as well, to improve the learning paradigm which could lead to the substitution of the human teacher necessary to set up the dictionary with a deeper and direct interaction between the SOM and the fuzzy classifier.

For the purpose of vegetation canopies' monitoring, the proposed methodology represents an alternative to the usage of MS data that, in this context, own a superior information content related to the possibility to operate in the near-infrared spectrum, which is particularly sensitive to chlorophyll. However, the well-known limitations related to weather conditions of passive sensors often hamper continuous surface monitoring. Some solutions, aiming at the combination of the positive characteristics of active and passive imagery (i.e., independence from weather and illumination conditions at one side and informative content at the other), were already proposed in the past literature (see [78], [79]) exploiting methodologies fusing the multifrequency information at the pixel level, ignoring the different semantics they carry. The proposed approach opens new feature-based fusion techniques combining SAR and MS information layers in the same fuzzy environment here discussed and to the semantic combination of archive and, for deforestation monitoring activities, to the semantic combination of multisensory images for the historical investigation of the changes occurring to the forest coverage.

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Donato Amitrano (Member, IEEE) was born in Naples, Italy, in December 27, 1985. He received the bachelor's degree in aerospace engineering, the master's degree in aerospace and astronautical engineering, and the Ph.D. degree in electronic and telecommunication engineering from the University of Naples Federico II, Naples, Italy, in 2009, 2012, and 2016, respectively.

From 2016 to April 2018, he was a Research Fellow with the University of Naples Federico II. From May 2018 to July 2020, he was a Research Associate with the Surrey Space Centre, University of Surrey, Guildford, U.K. He works as a Freelance Engineer and an Independent Researcher. In 2017, he was a co-founder and a CEO of Latitudo 40, Naples, a high-tech company spin-off of the University of Napoli Federico II, operating in the remote sensing sector. His main research interests are multitemporal synthetic aperture radar (SAR), remote-sensing techniques for developing countries, machine learning, human vision, SAR images interpretation and understanding, and data fusion.

Dr. Amitrano reached the second place (in team) in the University Challenge of the ESA Copernicus Masters competition in 2017.



Raffaella Guida (Member, IEEE) was born in Naples, Italy, in October 24, 1975. She received the Laurea degree *(cum laude)* in telecommunications engineering and the Ph.D. degree in electronic and telecommunications engineering from the University of Naples Federico II, Naples, Italy, in 2003 and 2007, respectively.

In 2006, she received a two-year research grant from the University of Naples Federico II to be spent at the Department of Electronic and Telecommunication Engineering on the topic of electromagnetic field propagation in urban environment. In 2006, she was also a Guest Scientist with the Department of Photogrammetry and Remote Sensing, Technische Universität München, Munich, Germany. In 2008, she joined the Surrey Space Centre (SSC), University of Surrey, Guildford, U.K., as a Lecturer of satellite remote sensing. She is still in SSC as a Reader, the Head of the Remote Sensing Applications research group, and the Director of the Postgraduate Research Program. Since 2008, she has coordinated, as PI or coI, national and international research projects on earth observation and has been awarded funding for a value of £2M. She has been involved in exchange and know-how transfer programs with Mexico, Colombia, Algeria, Nigeria, Spain, China, Philippines, and Qatar and has delivered bespoken courses on SAR and remote sensing to the Mexican Space Agency, Mexico City, Mexico, University Ecosur, Lerma Campeche, Mexico, University of Estado de Mexico, Toluca, Mexico, Universidad de Las Palmas de Gran Canaria, Las Palmas de Gran Canaria, Spain, Universidad del Rosario, Bogotá, Colombia, and China Academy of Space Technology, Beijing, China. Her main research interests are in the fields of electromagnetics and microwave remote sensing, particularly in simulation and modeling of synthetic aperture radar signals, new remote sensing mission concepts and applications, satellite data fusion, and interoperability.

Dr. Raffaella is one of the recipients of the IEEE JSTARS Best Paper Award 2013.



Pasquale Iervolino (Member, IEEE) was born in Naples, Italy, in August 16, 1985. He received the B.S. and M.S. degrees *(cum laude)* in telecommunications engineering from the University of Naples Federico II, Naples, Italy, in 2008 and 2010, respectively, and the Ph.D. degree in electronic engineering from the Surrey Space Centre, University of Surrey, Guildford, U.K., in 2016.

From 2015 to 2019, he had been working as a Research Fellow at the Surrey Space Centre on national and international research projects on earth observation. During his time at the Surrey Space Centre, he has been an Assistant Lecturer of the Satellite Remote Sensing and Electronic Laboratory courses and actively supported the design and the delivery of short courses on SAR, EO data, and its applications to the Mexican Space Agency, Mexico City, Mexico, University Ecosur, Lerma Campeche, Mexico, University of Estado de Mexico, Toluca, Mexico, and China Academy of Space Technology, Beijing, China. In September 2019, he has joined Airbus Defence and Space, Guildford, where he is developing novel applications and services from satellite products in the maritime domain and is involved in the ground segment operations for NovaSAR mission. His research interests include microwave remote sensing, the development and inversion of scattering models from natural and man-made surfaces, synthetic aperture radar ship-detection algorithms, and satellite imagery data fusion.

Dr. Iervolino received the Student Paper Award published on the IEEE PROCEEDINGS of the 2015 Asia–Pacific Conference on Synthetic Aperture Radar (APSAR).

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